

Problem

Rework Example 5.11 using the nonideal op amp LM324 instead of uA741

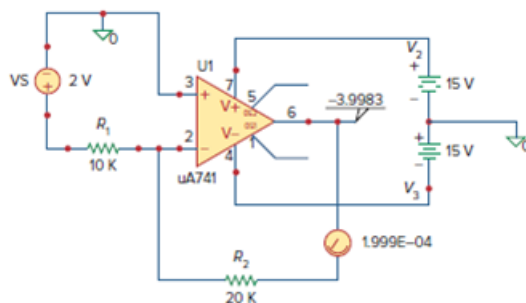
Example 5.11:

Use *PSpice* to solve the op amp circuit for Example 5.1.

Solution:

Using Schematics, we draw the circuit in Fig. 5.6(a) as shown in Fig. 5.35. Notice that the positive terminal of the voltage source v_s is connected to the inverting terminal (pin 2) via the 10-k Ω resistor, while the noninverting terminal (pin 3) is grounded as required in Fig. 5.6(a). Also, notice how the op amp is powered; the positive power supply terminal V_+ (pin 7) is connected to a 15-V dc voltage source, while the negative power supply terminal V_- (pin 4) is connected to -15 V. Pins 1 and 5 are left floating because they are used for offset null adjustment, which does not concern us in this chapter. Besides adding the dc power supplies to the original circuit in Fig. 5.6(a), we have also added pseudocomponents VIEWPOINT and IPROBE to respectively measure the output voltage v_o at pin 6 and the required current i through the 20-k Ω resistor.

Figure. 5.35: Schematic for Example 5.11.



After saving the schematic, we simulate the circuit by selecting **Analysis/Simulate** and have the results displayed on VIEWPOINT and IPROBE. From the results, the closed-loop gain is

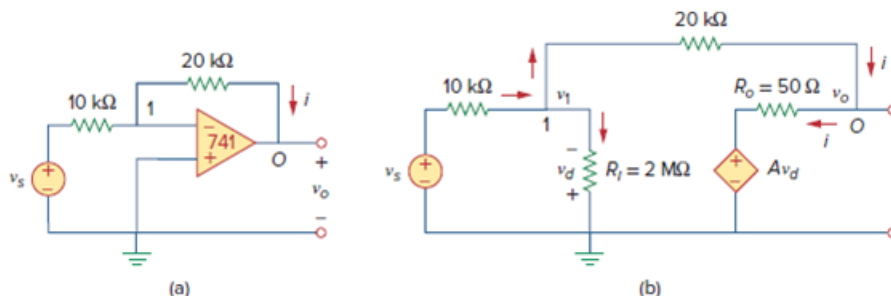
$$\frac{v_o}{v_s} = \frac{-3.9983}{2} = -1.99915$$

and $i = 0.1999$ mA, in agreement with the results obtained analytically in Example 5.1.

Example 5.1:

A 741 op amp has an open-loop voltage gain of 2×10^5 , input resistance of 2 M Ω , and output resistance of 50 Ω . The op amp is used in the circuit of Fig. 5.6(a). Find the closed-loop gain v_o/v_s . Determine current i when $v_s = 2$ V.

Figure. 5.6: (a) original circuit, (b) the equivalent circuit.

**Solution:**

Using the op amp model in Fig. 5.4, we obtain the equivalent circuit of Fig. 5.6(a) as shown in Fig. 5.6(b). We now solve the circuit in Fig. 5.6(b) by using nodal analysis. At node 1, KCL gives

$$\frac{v_s - v_1}{10 \times 10^3} = \frac{v_1}{2000 \times 10^3} + \frac{v_1 - v_o}{20 \times 10^3}$$

Multiplying through by 2000×10^3 , we obtain

$$200v_s = 301v_1 - 100v_o$$

Or

$$2v_s \approx 3v_1 - v_o \Rightarrow v_o = 2v_s + v_1 \quad (5.11)$$

At node O,

$$\frac{v_1 - v_o}{20 \times 10^3} = \frac{v_o - Av_d}{50}$$

But $v_d = -v_1$ and $A = 200,000$. Then

$$v_1 - v_o = 400(v_o + 200,000v_1) \quad (5.1.2)$$

Substituting v_1 from Eq. (5.1.1) into Eq. (5.1.2) gives

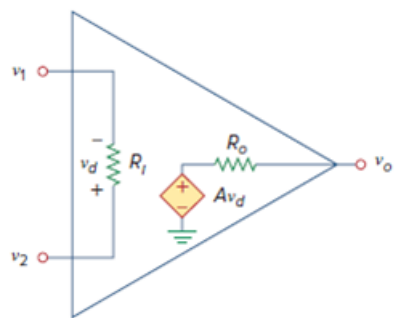
$$0 \simeq 26,667,067v_o + 53,333,333v_s \Rightarrow \frac{v_o}{v_s} = -1.9999699$$

This is closed-loop gain, because the 20-k Ω feedback resistor closes the loop between the output and input terminals. When $v_s = 2$ V, $v_o = -3.9999398$ V. From Eq. (5.1.1), we obtain $v_1 = 20.066667$ μ V. Thus,

$$i = \frac{v_1 - v_o}{20 \times 10^3} = 0.19999 \text{ mA}$$

It is evident that working with a nonideal op amp is tedious, as we are dealing with very large numbers.

Figure. 5.4: The equivalent circuit of the nonideal op amp.



Step-by-step solution

Step 1 of 3

Refer to figure 5.90 in the textbook.

As the voltage at the non-inverting terminals of both the operational amplifiers is 0 V, the voltage at the inverting terminals of the both the operational amplifiers is 0 V. Since, both the terminals of operational amplifier are virtually grounded.

Redraw the circuit as shown in figure 1.

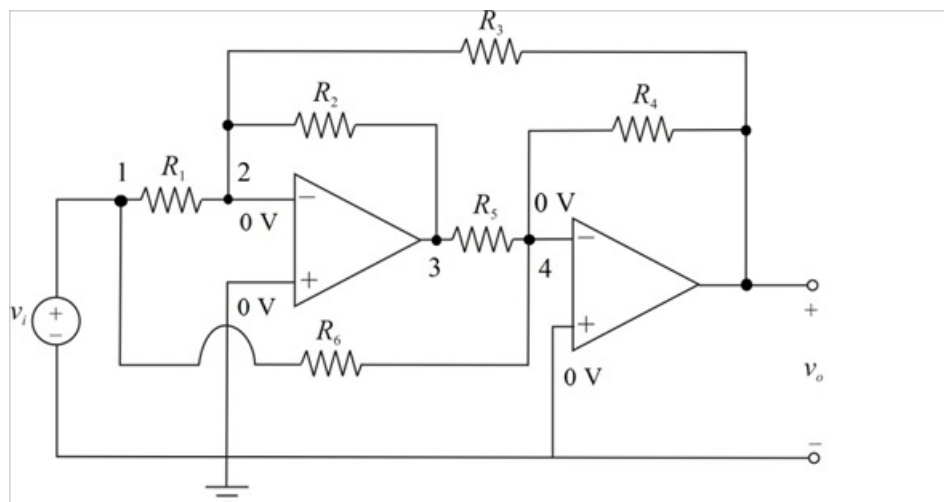


Figure 1

Step 2 of 3

Assume that voltage at node 3 is V_1 .

Apply Kirchhoff's current law at node 2 of the circuit shown in figure 1.

$$\frac{0 - v_i}{R_1} + \frac{0 - v_o}{R_3} + \frac{0 - V_1}{R_2} = 0 \dots\dots (1)$$

Apply Kirchhoff's current law at node 4 of the circuit shown in figure 1.

$$\frac{0 - V_1}{R_5} + \frac{0 - v_o}{R_4} + \frac{0 - v_i}{R_6} = 0$$

$$V_1 = -R_5 \left(\frac{v_o}{R_4} + \frac{v_i}{R_6} \right)$$

Step 3 of 3

Substitute $-R_5 \left(\frac{v_o}{R_4} + \frac{v_i}{R_6} \right)$ for V_1 in equation (1).

$$\frac{0 - v_i}{R_1} + \frac{0 - v_o}{R_3} + \frac{0 - \left[-R_5 \left(\frac{v_o}{R_4} + \frac{v_i}{R_6} \right) \right]}{R_2} = 0$$

$$\frac{0 - v_i}{R_1} + \frac{0 - v_o}{R_3} + \frac{R_5 \left(\frac{v_o}{R_4} + \frac{v_i}{R_6} \right)}{R_2} = 0$$

$$v_o \left(\frac{R_5}{R_2 R_4} - \frac{1}{R_3} \right) = v_i \left(\frac{1}{R_1} - \frac{R_5}{R_2 R_6} \right)$$

$$\frac{v_o}{v_i} = \frac{\frac{1}{R_1} - \frac{R_5}{R_2 R_6}}{\frac{R_5}{R_2 R_4} - \frac{1}{R_3}}$$

Simplify further,

$$\frac{v_o}{v_i} = \frac{\left(\frac{R_2 R_6 - R_1 R_5}{R_1 R_2 R_6} \right)}{\left(\frac{R_3 R_5 - R_2 R_4}{R_2 R_3 R_4} \right)}$$

$$\frac{v_o}{v_i} = \frac{R_3 R_4}{R_1 R_6} \left(\frac{R_2 R_6 - R_1 R_5}{R_3 R_5 - R_2 R_4} \right)$$

$$\frac{v_o}{v_i} = \frac{\frac{R_2 R_3 R_4}{R_1} - \frac{R_3 R_4 R_5}{R_6}}{R_3 R_5 - R_2 R_4}$$

Simplify further,

$$\frac{v_o}{v_i} = \frac{R_3 R_5 \left(\frac{R_2 R_4}{R_1 R_5} - \frac{R_4}{R_6} \right)}{R_3 R_5 - R_2 R_4}$$

$$\frac{v_o}{v_i} = \frac{\frac{R_2 R_4}{R_1 R_5} - \frac{R_4}{R_6}}{1 - \frac{R_2 R_4}{R_3 R_5}}$$

Therefore, the gain $\frac{v_o}{v_i}$ is

$$\boxed{\frac{\frac{R_2 R_4}{R_1 R_5} - \frac{R_4}{R_6}}{1 - \frac{R_2 R_4}{R_3 R_5}}}$$